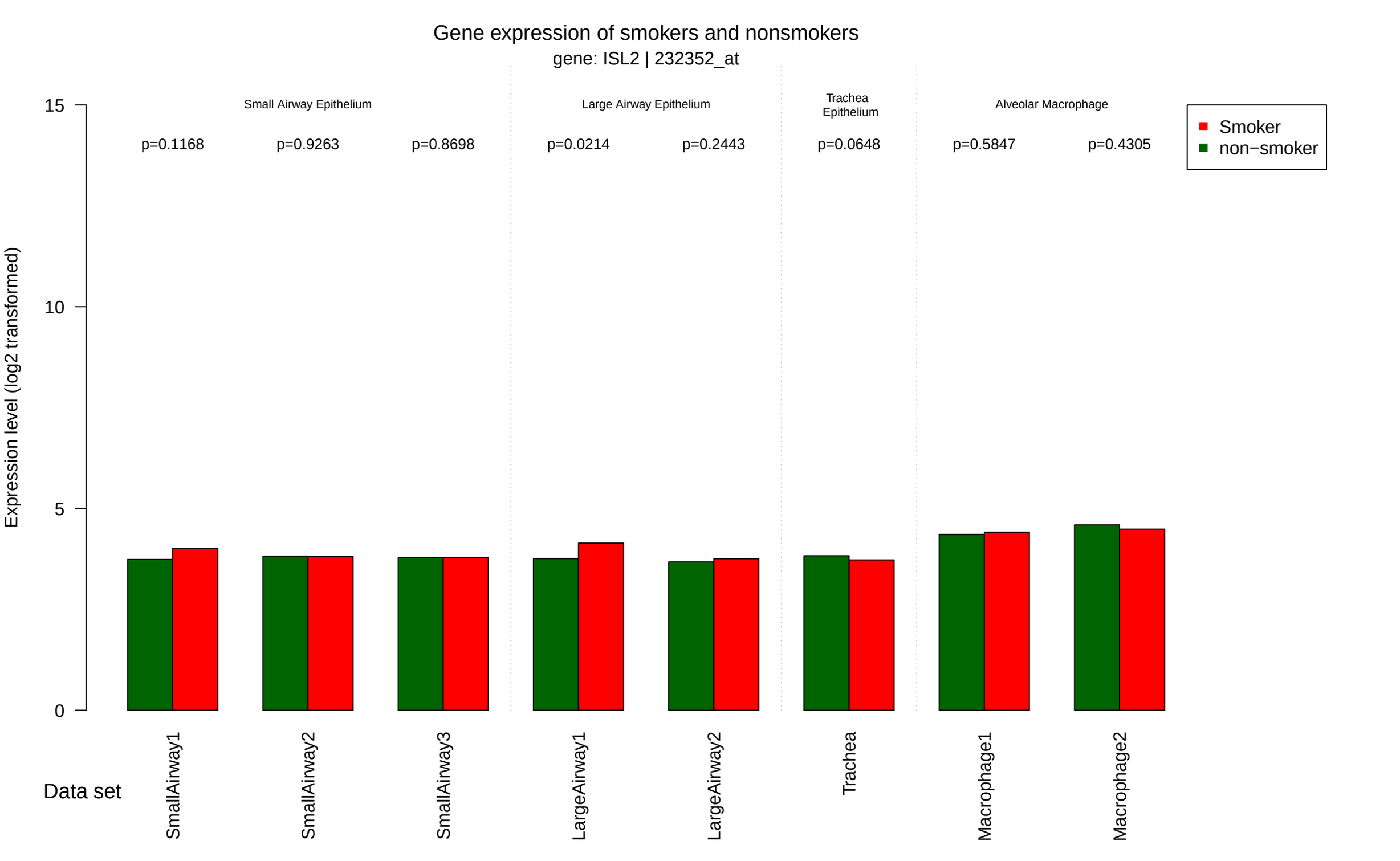
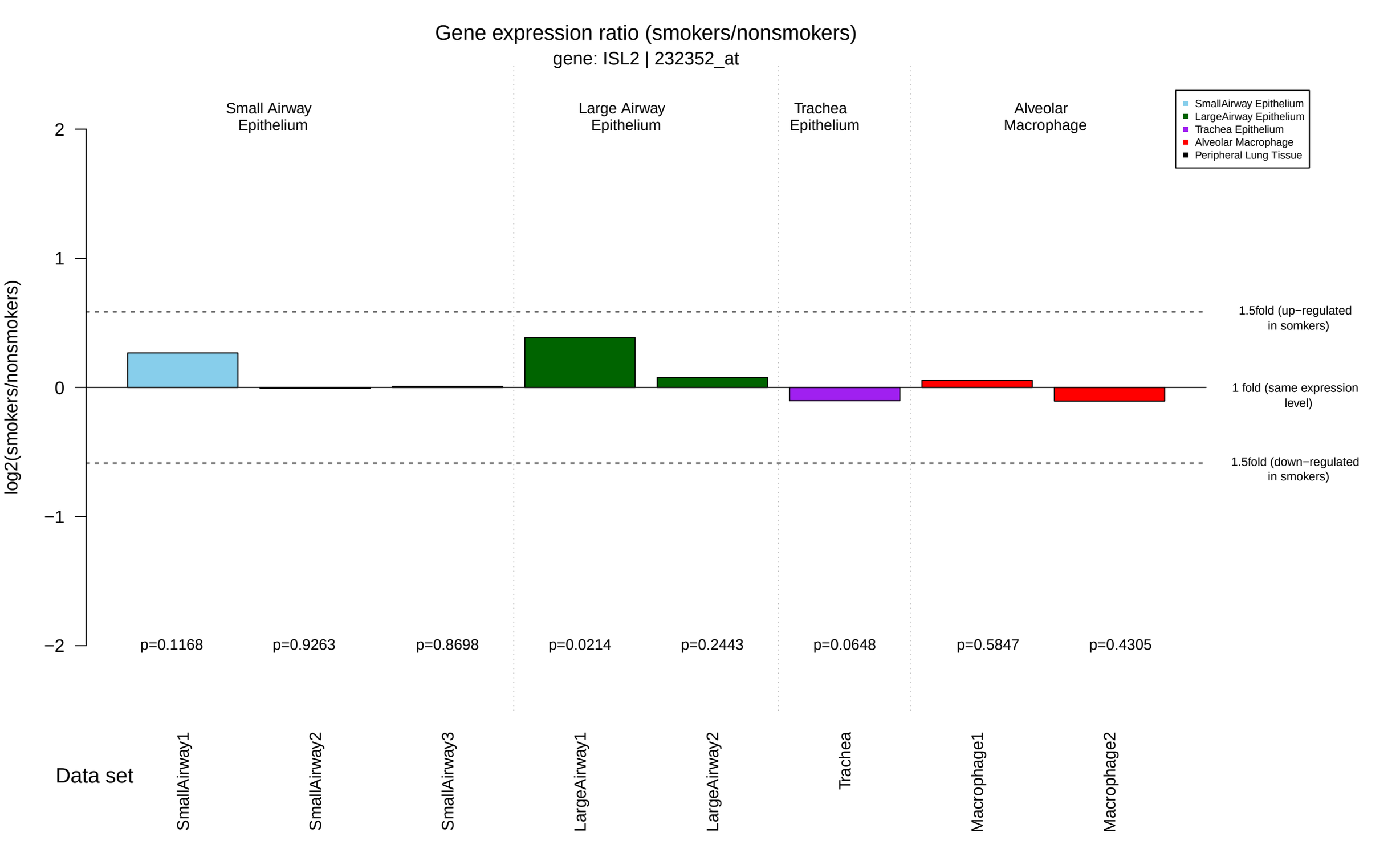
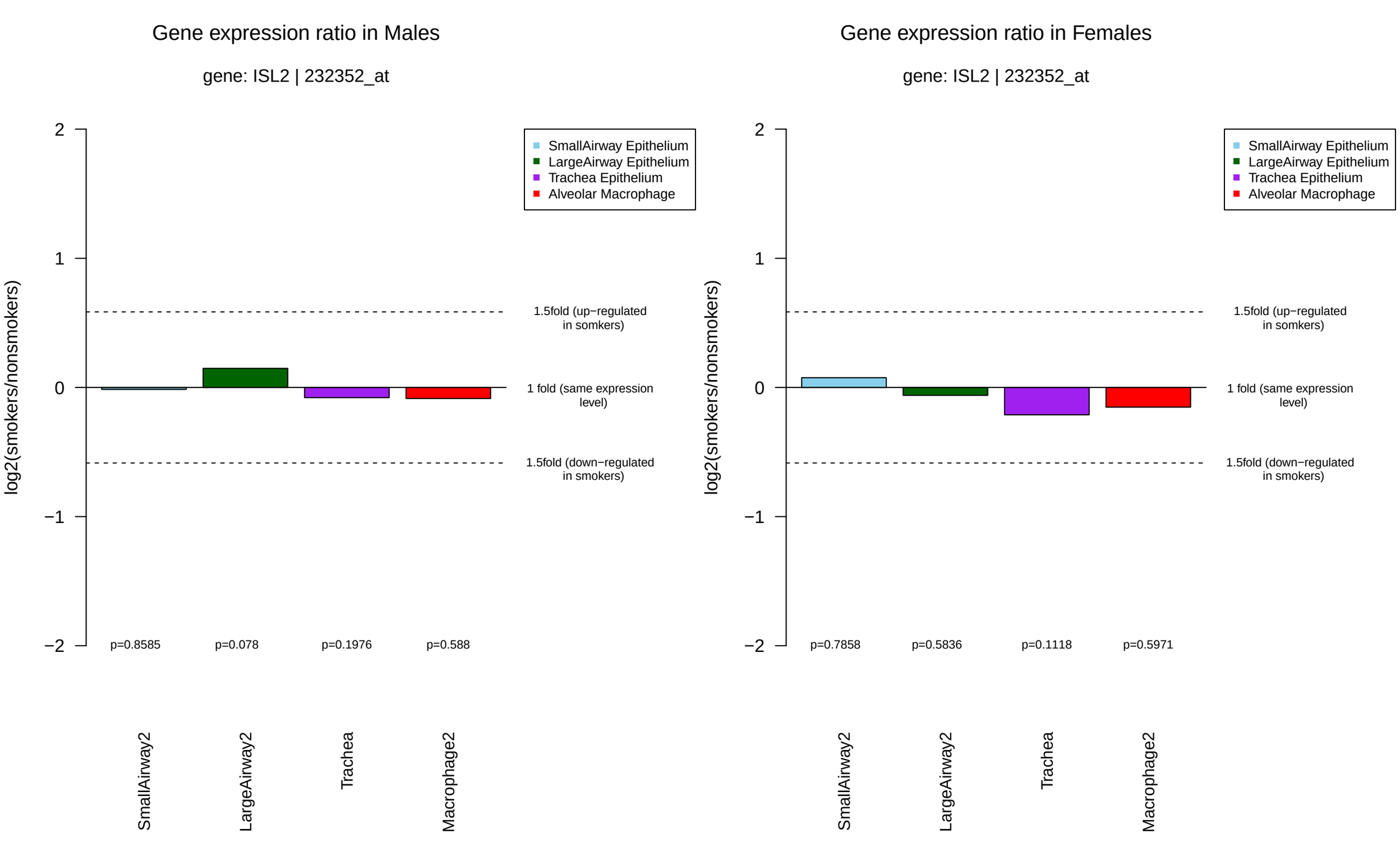
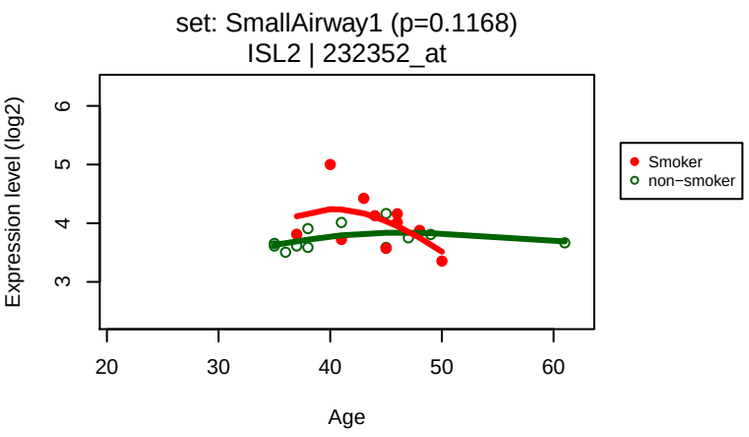
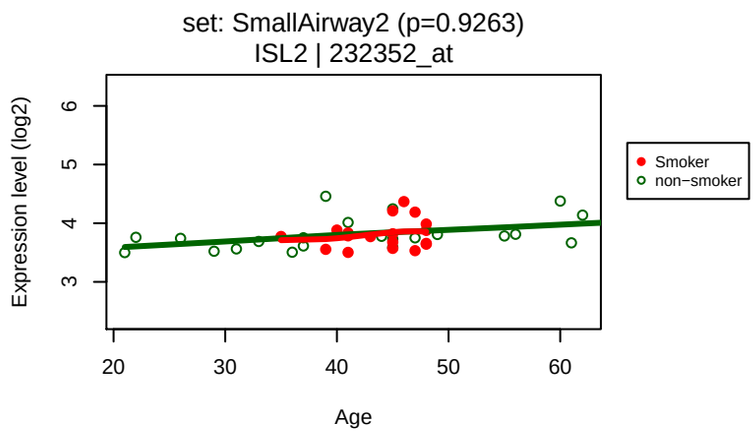
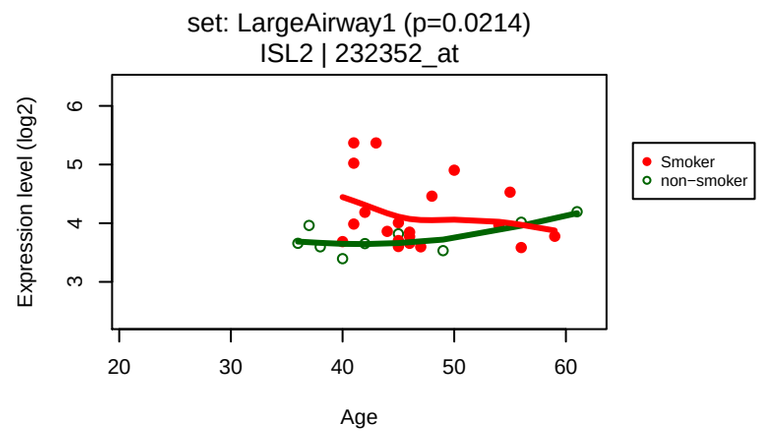

Gene expression of smokers and nonsmokers
gene: ISL2 | 232352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1168
p=0.9263
p=0.8698
p=0.0214
p=0.2443
p=0.0648
p=0.5847
p=0.4305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

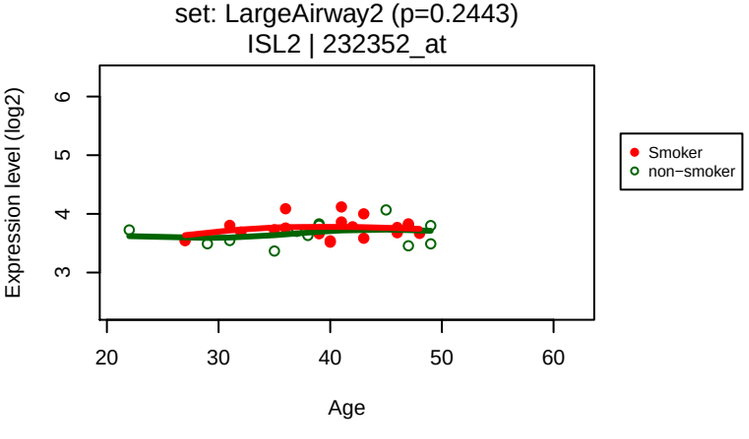

Gene expression ratio (smokers/nonsmokers)
gene: ISL2 | 232352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1168
p=0.9263
p=0.8698
p=0.0214
p=0.2443
p=0.0648
p=0.5847
p=0.4305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

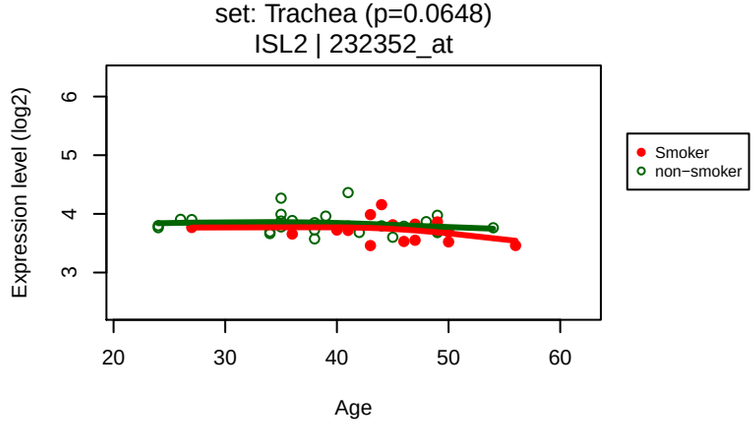

Gene expression ratio in Males
gene: ISL2 | 232352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8585
p=0.078
p=0.1976
p=0.588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISL2 | 232352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7858
p=0.5836
p=0.1118
p=0.5971
SmallAirway2
LargeAirway2
Trachea
Macrophage2

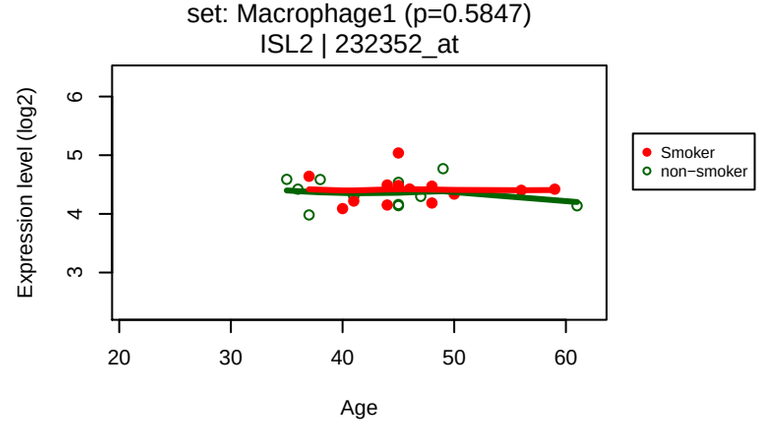

set: SmallAirway1 (p=0.1168)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker

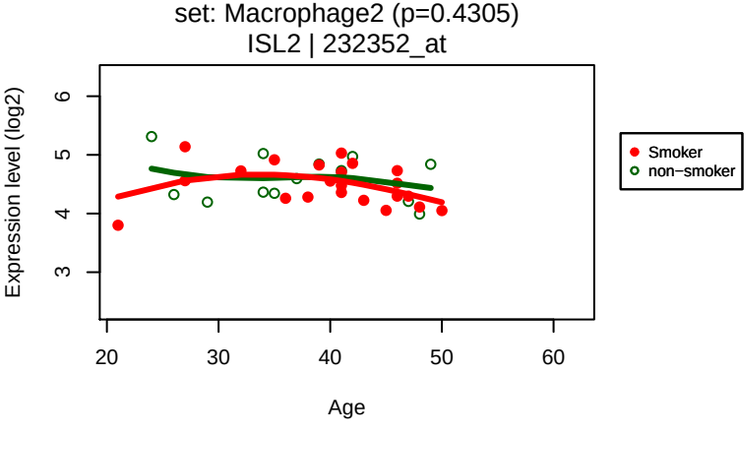

set: SmallAirway2 (p=0.9263)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker

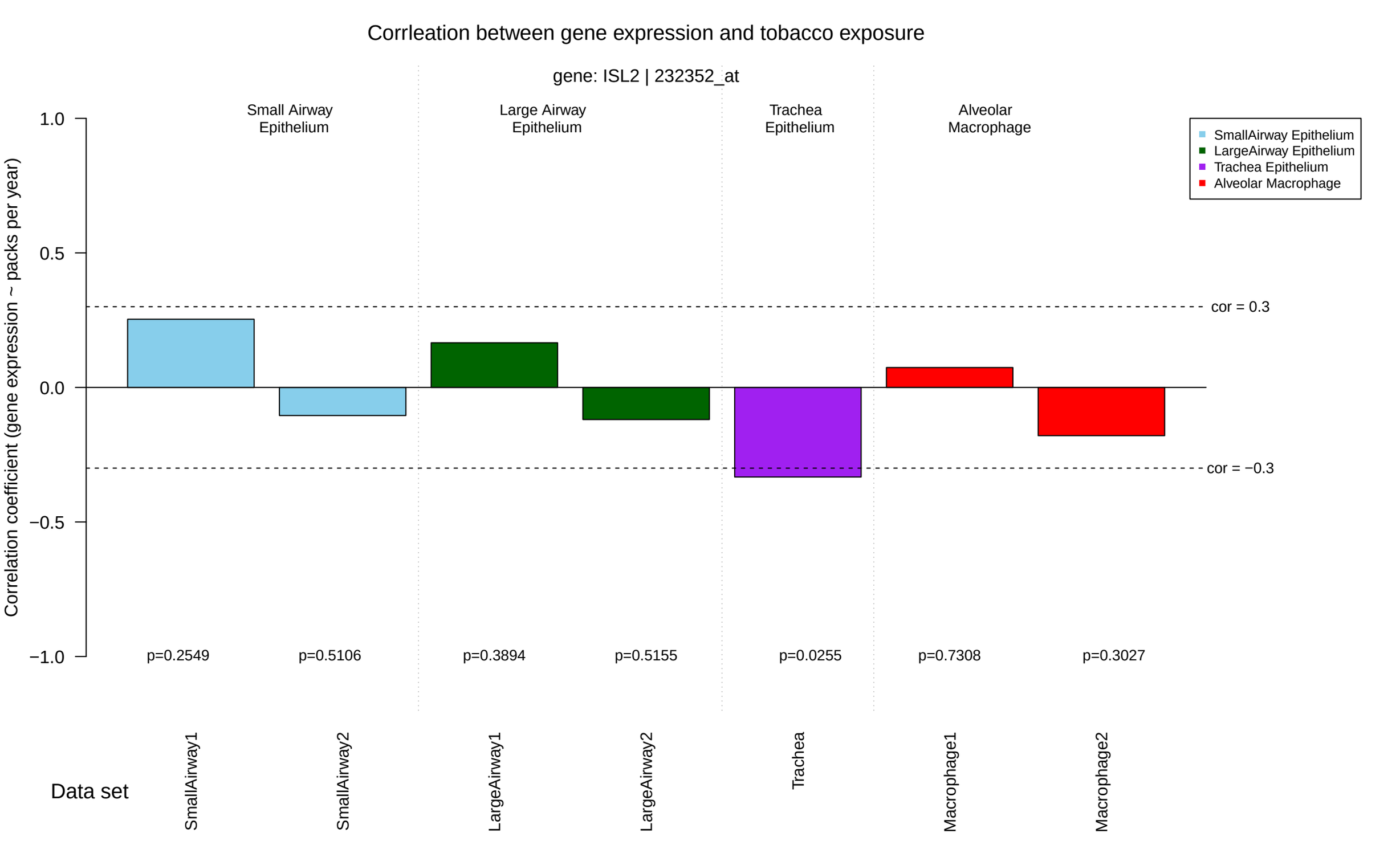

set: LargeAirway1 (p=0.0214)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2443)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0648)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5847)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4305)
ISL2 | 232352_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISL2 | 232352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2549
p=0.5106
p=0.3894
p=0.5155
p=0.0255
p=0.7308
p=0.3027
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2